

PART 1 — Why Model Evaluation Matters More Than Training

Chapter 14 — Model Evaluation Part 1 — Why Model Evaluation is More Important Than Model Training Before Starting Congratulations. Until now our project looks like this. **LTspice** — CSV Files.

17 Features

GridSearchCV — Best SVM. Prediction Question. Project finished? Most students say Yes. Wrong.

In Machine Learning,

training is only half the project. Question. How do we know whether our model is actually good? Need Evaluation.

What you will know after finishing this chapter

After finishing this chapter, you should be able to answer [OK] What is Model Evaluation? [OK] Why isn't training enough? [OK] Why isn't Accuracy enough? [OK] What is a Confusion Matrix? [OK] Why did Evaluation completely change our project?

Difficulty

Basics 5/5

Theory 5/5

Coding 2/5

Viva Importance 5/5

Let's Begin

Suppose I train an SVM. Question. How do I know whether it learned correctly? Simple. We compare its predictions with the actual labels. Example **Actual** — RC_Open. **Prediction** — RC_Open. Correct. Another sample. **Actual** — RC_Open. **Prediction** — RC_Normal. Wrong. Now count how many are correct and how many are wrong. That process is called Model Evaluation.

PROFESSOR ASKS

What is Model Evaluation?

EXCELLENT ANSWER

Model evaluation is the process of measuring how accurately a trained machine learning model predicts unseen data using suitable performance metrics. Remember the words unseen data. Very important. Why Not Evaluate on Training Data? Question. Suppose the model correctly predicts every training sample. Question. Can we conclude the model is excellent? No. Because it has already seen those samples. Real evaluation must always use the test dataset.

Real Life Analogy

Imagine studying 100 questions. During the exam, your teacher asks those exact 100 questions. Question. Will you score well? Probably yes. Question. Does that prove deep understanding? No. Need new questions. Exactly why

Machine Learning

uses a test set. The Simplest Metric Accuracy Question. What is Accuracy? Simple. Accuracy is the percentage of predictions that are correct. Suppose our model tests 100 samples. **Correct predictions** — 95. **Wrong predictions** — 5. Question. Accuracy? $95 / 100 = 95\%$ Very simple.

PROFESSOR ASKS

What is Accuracy?

EXCELLENT ANSWER

Accuracy is the proportion of correctly classified samples among the total number of samples evaluated. Is High Accuracy Always Good? Most students say Yes. Wrong. Let's see why. Imagine 100 patients. 99 Healthy 1 Cancer Now imagine a terrible model. It predicts Everyone Healthy Question. Accuracy? 99% Question. Is this a good model? Absolutely not. It missed the most important case. Very important. Accuracy can be misleading.

PROFESSOR ASKS

Why is Accuracy not enough?

EXCELLENT ANSWER

Accuracy alone can be misleading, especially when some classes are much more important or when the dataset is imbalanced. Additional evaluation metrics are required for a complete assessment. Excellent.

Our Project

Question. Did we only look at Accuracy? No. Very important. We examined individual fault classes. Question. Why? Because one fault was causing most of the errors. Remember RC_Open Exactly. Accuracy did not tell us that story. The

Confusion Matrix

did.

The Confusion Matrix

Now the most important evaluation tool in our project. Question. What is a Confusion Matrix? Simple. It is a table that compares

Actual Labels

with Predicted Labels. Example Predicted

A B

Actual A 45 5

Actual B 2 48

Notice every prediction is recorded. Nothing is hidden.

PROFESSOR ASKS

What is a Confusion Matrix?

EXCELLENT ANSWER

A confusion matrix is a table that compares the actual class labels with the predicted class labels, allowing detailed analysis of classification performance for every class. Excellent. Understanding the Matrix Question. What does this value mean? **Actual RC_Open** — Predicted RC_Open. Correct prediction. Good. Question. What about **Actual RC_Open** — Predicted RC_Normal. Wrong prediction. This tells us exactly which classes are being confused. Why Is This Powerful? Question. Suppose Accuracy is 96%. Question. Which fault is failing? Accuracy cannot tell us.

Confusion Matrix

can. This is exactly how our investigation began.

The Detective Story Begins

This is one of my favourite parts of the project. Question. How did we discover the problem? Not by looking at Accuracy. Instead, we looked at the Confusion Matrix. Question. What did we notice? One class kept appearing in the wrong place. That class was RC_Open This started the entire investigation.

Engineering Mindset

Notice our thinking. We did NOT say Accuracy 96% Looks good. Done. Instead we asked Why is RC_Open being confused? That question changed the entire project.

This Is Real Research

Many students stop when accuracy is high. Researchers start when they see an error. Remember this. It is one of the biggest lessons of this project.

PROFESSOR ASKS

Why did you analyze the confusion matrix even though the accuracy was already high?

EXCELLENT ANSWER

Although the overall accuracy was high, the confusion matrix revealed that specific fault classes were still being misclassified. Analyzing these errors helped identify missing information in the feature set and led to improvements in the model. This is one of the strongest answers in your viva.

Connection to Our Project

Our pipeline now becomes

Train Model

Predict — Evaluate.

Confusion Matrix

Find Weaknesses

Improve Model

Notice Evaluation is not the end. Evaluation starts the improvement cycle.

COMMON MISTAKES

MISTAKE: [X] High Accuracy means perfect model. Wrong.

MISTAKE: [X] Training Accuracy and Test Accuracy are the same. Wrong.

MISTAKE: [X] Evaluation ends after calculating Accuracy. Wrong. Evaluation includes detailed error analysis.

REVISION NOTES

Remember these ideas.

Accuracy — Overall Performance.

Confusion Matrix — Class-wise Performance.

Evaluation — Find Errors.

Errors — Improve Model.

MEMORY TRICK

Remember this sentence. "Accuracy tells how good the model is. The Confusion Matrix tells why it is not perfect." One sentence. Entire chapter.

KNOWLEDGE CHECK

Level 1

- What is Model Evaluation?
- What is Accuracy?
- What is a Confusion Matrix?
- Why is the test set used for evaluation?
- Why isn't Accuracy alone sufficient?

Level 2

- Explain how a confusion matrix is constructed.
- Why can two models have the same Accuracy but different Confusion Matrices?
- Why is class-wise analysis important?
- How did the confusion matrix help improve our project?
- Why should evaluation lead to model improvement?

Professor Level

- Explain the complete evaluation pipeline used in your project.
- Why is a confusion matrix more informative than overall accuracy?
- Describe how the RC_Open issue was first identified.
- Why is error analysis an essential part of machine learning research?
- How does model evaluation guide feature engineering?
- End of Chapter 14 (Part 1)
- Congratulations.
- You now understand why evaluation is not just about reporting numbers—it is about understanding the model's behavior.
- This chapter marks the point where your project transitions from building a classifier to improving a classifier through systematic analysis.

Author's Note The next part is one of the most important sections in the entire Project Bible. We'll study: True Positive (TP) True Negative (TN) False Positive (FP) False Negative (FN) Precision Recall F1-Score Classification Report These metrics are the foundation for understanding the detailed performance of your classifier and are commonly asked in placements, interviews, and vivas. After that, we'll return to your project's actual confusion matrix and begin the RC_Open detective story, which ultimately led to DC Level and the Topology Rule.

PART 2 — TP, TN, FP, FN, Precision, Recall & F1-Score

Chapter 14 — Model Evaluation Part 2 — TP, TN, FP, FN, Precision, Recall & F1-Score Before Starting

In Part 1

we learned that Accuracy is not enough. Question. Why? Because Accuracy cannot tell us which classes are making mistakes. Need more detailed evaluation. That is exactly what we'll learn now.

First Question

Suppose our SVM predicts one fault. Question. Can the prediction be correct or wrong? Yes. Question. Can the actual class also be correct or wrong? Yes. Combining these gives four possibilities. These are the foundation of Machine Learning Evaluation.

The Four Cases

They are

True Positive

(TP) True Negative (TN) False Positive (FP) False Negative (FN) Every evaluation metric comes from these four values. Let's Understand Using Our Project Suppose we are checking only one class. Example RC_Open Think of it as YES or NO. Question. Is this RC_Open? Case 1 True Positive (TP) Suppose **Actual** — RC_Open. **Prediction** — RC_Open. Question. Correct? Yes. Question. Positive? Yes. Therefore True Positive.

PROFESSOR ASKS

What is True Positive?

EXCELLENT ANSWER

A True Positive occurs when the model correctly predicts a positive sample as positive.

Real Life Analogy

Doctor tests a patient. Patient actually has a disease. Doctor correctly diagnoses the disease. That is True Positive. Case 2 True Negative (TN) Suppose **Actual** — Not RC_Open. **Prediction** — Not RC_Open. Correct again. Negative. Therefore True Negative.

PROFESSOR ASKS

What is True Negative?

EXCELLENT ANSWER

A True Negative occurs when the model correctly predicts a negative sample as negative.

Real Life Analogy

Healthy patient. Doctor says Healthy. Correct. True Negative. Case 3 False Positive (FP) Now comes an important mistake. **Actual** — RC_Normal. **Prediction** — RC_Open. Question. Was the prediction correct? No. Question. Did the model predict Positive? Yes. Therefore False Positive.

PROFESSOR ASKS

What is False Positive?

EXCELLENT ANSWER

A False Positive occurs when the model incorrectly predicts a negative sample as positive.

Real Life Analogy

Airport Security. Alarm rings. Everyone panics. Later they discover there was no bomb. False Alarm. False Positive. Very easy to remember. Case 4 False Negative (FN) Now the opposite. **Actual** — RC_Open. **Prediction** — RC_Normal. Question. Prediction correct? No. Question. Actual was Positive? Yes. Question. Model missed it? Yes. False Negative.

PROFESSOR ASKS
What is False Negative?
EXCELLENT ANSWER

A False Negative occurs when the model incorrectly predicts a positive sample as negative.

Real Life Analogy

Patient has cancer. Doctor says Healthy. Very dangerous. False Negative.

Easy Memory Trick

Remember this table.

Positive	Positive	TP
Negative	Negative	TN
Negative	Positive	FP
Positive	Negative	FN

Memorize this table. It appears everywhere in Machine Learning. Precision Now we build our first metric. Question. Suppose our model predicts RC_Open 100 times. Question. How many were actually RC_Open? That is Precision. Simple.

PROFESSOR ASKS
What is Precision?
EXCELLENT ANSWER

Precision measures the proportion of positive predictions that are actually correct.

Real Life Analogy

Suppose a teacher selects 10 students for a scholarship. Question. How many actually deserved the scholarship? That is Precision.

High Precision Means

When our model says RC_Open it is usually correct. Very few False Positives. Recall Now another metric. Question. Suppose there are 100 actual RC_Open circuits. Question. How many did our model find? That is Recall.

PROFESSOR ASKS
What is Recall?
EXCELLENT ANSWER

Recall measures the proportion of actual positive samples that are correctly identified by the model.

Real Life Analogy

Imagine a police officer trying to catch all criminals. Question. Did he catch everyone? That is Recall. Difference Between Precision and Recall Students always confuse these. Remember this. Precision asks When I predict positive, how often am I correct? Recall asks Out of all actual positives, how many did I find? Completely different.

F1 Score

Question. Suppose Precision is excellent. Recall is poor. Question. Is the model good? Not necessarily. Need one metric that balances both. That is F1 Score.

PROFESSOR ASKS

What is F1 Score?

EXCELLENT ANSWER

The F1 Score is the harmonic mean of Precision and Recall and provides a balanced measure when both metrics are important.

Real Life Analogy

Imagine a student. **Math** — 100. **Physics** — 20. Question. Overall excellent? Not really. Need balance.

F1 Score

does exactly that.

Classification Report

Question. Suppose our project contains many classes. RC_Normal RC_Open RL_Normal RL_Open. Question. Will we calculate Precision, Recall and F1 for every class? Yes. Scikit-learn collects everything inside one report. This is called the Classification Report.

PROFESSOR ASKS

What is a Classification Report?

EXCELLENT ANSWER

A Classification Report summarizes Precision, Recall, F1 Score, and Support for every class in a classification problem. Support Students often ignore Support. Question. What is Support? Simple. Support means the number of actual samples belonging to that class. Example **RC_Open** — 120 samples. Support = Nothing more.

Connection to Our Project

When our SVM finished prediction, our evaluation included Accuracy

Overall Performance

Precision

Prediction Quality

Recall

Detection Ability

F1 Score

Balanced Performance

Confusion Matrix

Class-wise Errors Together these gave us the complete picture of our classifier.

Why This Was Important

Question. Would Accuracy alone have told us that RC_Open was problematic? No. Question. Would the Classification Report and

Confusion Matrix

show that class had lower Recall or more misclassifications? Yes. Exactly. That is why we looked beyond Accuracy.

COMMON MISTAKES

MISTAKE: [X] Precision and Recall are the same. Wrong.

MISTAKE: [X] High Accuracy guarantees high Recall. Wrong.

MISTAKE: [X] F1 Score is just another name for Accuracy. Wrong.

MISTAKE: [X] Support measures accuracy. Wrong. Support only counts samples.

REVISION NOTES

Remember these ideas.

TP — Correct Positive.

TN — Correct Negative.

FP — False Alarm.

FN — Missed Detection.

Precision — How correct are positive predictions?

Recall — How many actual positives were found?

F1 — Balance between Precision and Recall.

Support — Number of samples.

MEMORY TRICK

Remember this sentence. "Precision trusts the predictions. Recall finds the positives. F1 balances both." One sentence. Entire chapter.

KNOWLEDGE CHECK

Level 1

- What is TP?
- What is FP?
- What is FN?
- What is Precision?
- What is Recall?

Level 2

- Explain the difference between Precision and Recall.
- Why do we need the F1 Score?
- What is Support?
- Why is the Classification Report useful?
- Why isn't Accuracy sufficient?

Professor Level

- Explain all four outcomes (TP, TN, FP, FN) using the RC_Open class from your project.
- Why is Precision important in fault diagnosis?
- Why can a model have high Precision but low Recall?
- Why is the F1 Score preferred when Precision and Recall must both be considered?
- Which evaluation metrics helped you identify weaknesses in your classifier?
- End of Chapter 14 (Part 2)
- Congratulations.
- You now understand the complete foundation of machine learning evaluation metrics.
- These concepts are asked in almost every ML interview, GATE DA viva, and project presentation.

Author's Note The next continuation is one of the most exciting parts of the entire Project Bible. We'll stop talking about generic machine learning and return completely to your project. We'll analyze: Your actual confusion matrix Why RC_Open kept getting misclassified The complete RC_Open Detective Story Every hypothesis we tested Why some hypotheses failed Why DC Level was introduced Why DC Level alone was not enough And how this ultimately led to the Topology Rule, which is arguably the most original engineering contribution in your project. This next section is where your project changes from using machine learning to doing engineering research. It is also the part that will make your viva stand out because it shows your reasoning process, not just your final results.

PART 3 — The RC_Open Detective Story

Chapter 14 — Model Evaluation Part 3 — The RC_Open Detective Story (The Turning Point of Our Project) Before Starting Everything we learned until now was preparation. Now we reach the part that transformed our project from a normal Machine Learning project into an engineering project. This chapter is not about algorithms. It is about thinking.

The Situation

Our project was finished. We had ✓ LTspice

✓ Dataset

✓ Feature Extraction

✓ SVM ✓ GridSearchCV

✓ High Accuracy

Everything looked good. Question. Should we submit it? Most students would. We didn't.

The First Suspicion

Instead of looking only at Accuracy, we opened the Confusion Matrix. Question. Why? Because we wanted to know which classes were making mistakes. Not just how many. When we looked carefully, one class immediately caught our attention. RC_Open Again. Again. Again. It kept appearing in wrong predictions. Question. Why only RC_Open? That became our research question.

PROFESSOR ASKS

How did you discover the RC_Open problem?

EXCELLENT ANSWER

The problem was discovered during confusion matrix analysis. Although the overall accuracy was high, RC_Open showed repeated misclassification, indicating that the classifier was struggling to distinguish that particular fault class. This answer is much stronger than simply saying "Accuracy was low."

Step 1 — We Did NOT Change the Model

This is extremely important. Many students immediately say Let's use Deep Learning. Or Let's use Random Forest. Or Let's increase epochs. Question. Did we do that? No. Instead we asked a much better question. Why is SVM making this mistake? Notice the difference. One approach changes the algorithm. The other tries to understand the problem. Real engineers always choose the second approach.

Step 2 — Compare the Waveforms

Question. What should we compare? Simple. Take Normal RC. Take RC_Open. Place them side by side. Observe. Question. Can we see a major difference? Some differences were obvious. Some were not. Then we looked at the extracted features.

Step 3 — Compare the Features

Question.

Was Peak

very different? Not always. Question. Was RMS very different? Sometimes. Sometimes not. Question.

Was Mean

enough? Again not always. Question. What about Rise Time? Settling Time? Energy? Frequency? Still not enough. Now something became clear. We had many features. Yet the classifier still confused RC_Open. Question. What information was missing? This Is the Most Important Question Not "What

algorithm should we use?" Instead "What information does the classifier not know?" That question changed the project.

Engineering Thinking

Imagine a detective. He doesn't say "I'll arrest another person." He asks "What clue am I missing?" Exactly the same idea.

Machine Learning

had clues. But one clue was missing.

Step 4 — Looking at the Steady-State

We looked at the waveforms again. This time more carefully. Question. After the transient finished, what happened? The waveform settled. Question. Was that final value important? Yes. Very important. Then we realised something shocking. None of our features explicitly described that steady-state value. We had Mean. But Mean uses the entire waveform. We had Peak. We had RMS. We had Energy. But none of them specifically answered one simple question. What voltage remains after everything settles?

The Missing Clue

That steady-state value was the missing clue. It was not explicitly present inside the feature vector. Now the solution became obvious. Create one new feature. Birth of Feature 17 That feature was DC Level Question. Why DC Level? Because it directly measures the steady-state voltage after the transient response has ended. Exactly the information we were missing.

PROFESSOR ASKS
Why did you introduce DC Level?
EXCELLENT ANSWER

Error analysis showed that the existing features did not explicitly represent the steady-state voltage. Since RC_Open exhibited a distinguishable steady-state behavior, DC Level was introduced to capture this missing information and improve fault classification. This is probably one of the strongest answers in your entire viva.

Step 5 — Retraining the Model

Now we added

Feature 17.

Question. Did we change the SVM? No. Question. Did we change GridSearchCV? No. Question. What changed? Only the information given to the classifier. Remember this. Sometimes better features are more powerful than better algorithms. What Happened Next? Question. Did DC Level improve the model? Yes. It definitely helped. But did it eliminate every error? No. Some confusion still remained. Question. Why? Now we reached an even deeper engineering problem.

A Bigger Discovery

The remaining problem was not Machine Learning. The remaining problem was Circuit Physics. This is where our project became much more interesting.

We Asked Another Question

Suppose two circuits have similar feature values. Question. Can they still belong to different topologies? Yes. Exactly. Now we realised something.

Machine Learning

cannot learn information that is not present in the features. Some ambiguity came from the circuit topology itself. That led us to our next idea.

The Topology Rule

We won't study it today. But remember this sequence.

Confusion Matrix

RC_Open Errors

Feature Investigation

Missing Information

DC Level

Remaining Errors

Topology Rule

This is the complete research journey of the project.

Why This Chapter Is So Important

Most student projects end here.

Train Model

Report Accuracy

Our project continued.

Train Model

Evaluate — Investigate

Understand — Improve

Evaluate Again

That is real engineering.

Connection to Our Project

Notice the complete cycle. Prediction

Confusion Matrix

RC_Open Problem

Engineering Investigation

New Feature

Better Classifier

This cycle is exactly what happened during our project development.

COMMON MISTAKES

MISTAKE: [X] Low performance always means change the algorithm. Wrong. Sometimes the features are the problem.

MISTAKE: [X] Accuracy is enough. Wrong. The confusion matrix revealed the real issue.

MISTAKE: [X] Feature engineering ends before training. Wrong. Feature engineering often continues after evaluation.

REVISION NOTES

Remember
the investigation.

Confusion Matrix — Found RC_Open issue.

Feature Analysis — Looked for missing information.

DC Level — Added missing steady-state information.

Remaining Errors — Led to Topology Rule.

MEMORY TRICK

Remember this sentence. "The confusion matrix found the problem. Feature engineering solved part of it. Circuit understanding solved the rest." That single sentence summarizes the evolution of your project.

KNOWLEDGE CHECK

Level 1

- Which fault class first attracted attention?
- Why didn't we immediately change the classifier?
- What information was missing from the feature vector?
- Which new feature was added?
- Did DC Level completely solve the problem?

Level 2

- Explain the complete reasoning process that led to DC Level.
- Why is error analysis important in feature engineering?
- Why did adding a new feature improve the classifier without changing the algorithm?
- Why were some errors still present after adding DC Level?
- What lesson does this investigation teach about machine learning projects?

Professor Level

- Describe the complete RC_Open investigation from confusion matrix to feature engineering.
- Why is feature engineering often more valuable than changing classifiers?
- Explain how confusion matrix analysis guided the evolution of your project.
- Why can two different circuit topologies still produce similar feature vectors?
- What does this investigation demonstrate about the relationship between machine learning and domain knowledge?
- End of Chapter 14 (Part 3)
- Congratulations.
- This chapter captures the most original reasoning process in your entire project.
- Anyone can train an SVM.
- Far fewer people can explain how they used the confusion matrix to discover a weakness, investigated the cause, introduced a new feature based on circuit behavior, and then continued investigating instead of stopping.
- That engineering mindset is one of the strongest aspects of your project.

Author's Note The next chapter will cover the Topology Rule in complete detail. This is arguably the most unique contribution in your project because it goes beyond standard machine learning. We'll explain: What the Topology Rule is. Why DC Level alone couldn't solve every ambiguity. How circuit topology affects feature similarity. Why two different faults can still confuse a classifier. How the Topology Rule complements machine learning instead of replacing it. How to explain this idea confidently in a viva, interview, or project presentation. This is the point where your project transitions from "using machine learning" to combining machine learning with engineering expertise, which is what makes it stand out.